You decide to check another of your assumptions:

> You: OK, how about "aBcdEfg1"?
>
> Raj: That one would actually be pretty easy for a machine to crack—it's just a sequence of alphabetically ordered letters and a number. Sequences are easy to crack, and just adding a single number at the end doesn't add much complexity.
>
> You: What about qwertY12
>
> Raj: That's just a sequence of keys on the keyboard. Most hacking algorithms know about that trick, so it would be very easy to guess.
>
> You: Oh. OK, how about "dJeZDip1"?
>
> Raj: That would be a bit short, but OK.

So now you have another requirement: alphabetical sequences of letters or spatial sequences of keys on the keyboard are both a no-no. But you've noticed something interesting:

> You: Raj, rather than just saying if a password I give you is secure or not, you seem to be grading them by how hard they are to hack—is that intentional?
>
> Raj: Well, I wasn't thinking of it like that, but yes, of course: the whole point of a secure password is so that it doesn't get hacked, and the passwords most people use are pretty easy to hack.[2] There are a lot of studies and a lot of algorithms out there that measure password strength.[3] And many sites provide feedback on the strength of the passwords you enter. In those terms, we need passwords to be of at least medium strength.
>
> Raj brings up a screen similar to the one in figure 4.15.
>
> You: Raj, I think we've been focusing on the detailed rules for password validation too much. Working through these examples seems to indicate that the rules are less clear-cut than we initially thought. And the rules we're describing focus on one particular solution to the problem we're trying to solve: the real value in this story comes from ensuring that users have a strong password, not enforcing a particular set of rules. If we reason in terms of password strength rather than specific rules, maybe we could rephrase the acceptance criteria like this:
>
> – The password should be at least of medium strength to be accepted.
> – I should be informed of the strength of my proposed password.
> – If the password is too weak, I should be informed why.
>
> Raj: Yes, that sounds fine. But how do we know what qualifies as a medium-strength password? (See figure 4.16.)

---

[2] See, for example, Dan Goodin, "Anatomy of a hack: even your 'complicated' password is easy to crack," http://www.wired.co.uk/news/archive/2013-05/28/password-cracking.

[3] For anyone interested in this field, there's an interesting article on password strength by Dan Wheeler, "zxcvbn: realistic password strength estimation," at https://tech.dropbox.com/2012/04/zxcvbn-realistic-password-strength-estimation.